Rwanda

Overview of foundational documents

Maize flour ✔

Mandatory fortification since 2019

Source: Rwanda Food and Drugs Authority. Regulations No CBD/TRG/003 Rev. No 1 governing Food Fortification in Rwanda, (Rwanda FDA law No. 003/2018 of 09/02/2019, Article 8). Rwanda. 14/January/2020. [http://www.rwandafda.gov.rw/web/fileadmin/RegulationsGoverningFoodFortification.pdf]

12 Countries in Africa have legislation for mandatory fortification of maize flour

Legislation scope for maize flour in Rwanda	
Type of Maize flour that must be fortified	All types (no exceptions)
Origins or destinations of Maize flour that must be fortified	✔ Domestically produced ✔ Imported ✘ Exports
Intended use of Maize flour that must be fortified	✔ Household ✔ Processed Food ✔ Animal Feed ✘ Donated food

Source: Rwanda Food and Drugs Authority. Regulations No CBD/TRG/003 Rev. No 1 governing Food Fortification in Rwanda, (Rwanda FDA law No. 003/2018 of 09/02/2019, Article 8). Rwanda. 14/January/2020. [http://www.rwandafda.gov.rw/web/fileadmin/RegulationsGoverningFoodFortification.pdf]

Nutrient levels and compounds in maize flour fortification standard in Rwanda		
B6	Pyridoxine There are both WHO recommended and non-WHO recommended compounds	5 mg/kg
B12	Cyanocobalamin All compounds are WHO recommended	0.015 mg/kg 38% of the WHO recommended level
Folate (B9)	Folic acid All compounds are WHO recommended	1.2 mg/kg 24% of the WHO recommended level
Iron	NaFeEDTA All compounds are WHO recommended	20 mg/kg 50% of the WHO recommended level
Niacin (B3)	Niacinamide All compounds are WHO recommended	30 mg/kg 83% of the WHO recommended level
Riboflavin (B2)	Riboflavin All compounds are WHO recommended	4 mg/kg 200% of the WHO recommended level
Thiamin (B1)	Thiamin mononitrate There are both WHO recommended and non-WHO recommended compounds	6.5 mg/kg
Vitamin A	Retinyl palmitate All compounds are WHO recommended	1 mg/kg 17% of the WHO recommended level
Zinc	Zinc oxide All compounds are WHO recommended	49 mg/kg 52% of the WHO recommended level

Source: Rwanda Bureau of Standards. Rwanda Standard, RS EAS 768: 2012, First edition, Published by RBS 2012, Fortified milled maize (corn) products-specification, Reference number RS EAS 768: 2012. Rwanda. 2012.

13 Countries in Africa have maize flour fortification standards

Presence of monitoring protocols for maize flour fortification in Rwanda	
External monitoring of domestic production	Unknown
Imported food monitoring	Unknown

Source for external monitoring protocols: Not applicable
Source for import monitoring protocols: Not applicable

7 Countries in Africa with mandatory fortification of maize flour have external monitoring protocols

3 Countries in Africa with mandatory fortification of maize flour have import monitoring protocols

Oil ✔

Mandatory fortification since 2020

Source: Rwanda Food and Drugs Authority. Regulations CBD/TRG/003 Rev. No. 1 Governing Food Fortification in Rwanda. Rwanda. 14/January/2020.

26 Countries in Africa have legislation for mandatory fortification of oil

Legislation scope for oil in Rwanda	
Type of Oil that must be fortified	All types (no exceptions)
Origins or destinations of Oil that must be fortified	✘ Domestically produced ✔ Imported ✘ Exports
Intended use of Oil that must be fortified	✔ Household ✔ Processed Food ✘ Animal Feed ✔ Donated food

Source: Rwanda Food and Drugs Authority. Regulations CBD/TRG/003 Rev. No. 1 Governing Food Fortification in Rwanda. 09/February/2019.

Nutrient levels and compounds in oil fortification standard in Rwanda		
Vitamin A	Retinyl palmitate	30 mg/kg

Source: Rwanda Food and Drugs Authority. Regulations CBD/TRG/003 Rev. No. 1 Governing Food Fortification in Rwanda. Rwanda. 14/January/2020.

28 Countries in Africa have oil fortification standards

Presence of monitoring protocols for oil fortification in Rwanda	
External monitoring of domestic production	Not applicable
Imported food monitoring	Not applicable

Source for external monitoring protocols: Not applicable
Source for import monitoring protocols: Not applicable

12 Countries in Africa with mandatory fortification of oil have external monitoring protocols

1 Country in Africa with mandatory fortification of oil have import monitoring protocols

Rice

Unknown fortification status

Salt

Mandatory fortification since 2019

Source: Rwanda Food and Drugs Authority. Regulations No. CBD/TRG/003 Rev No. 1 Governing Food Fortification in Rwanda. Kigali, Rwanda. 14/January/2020

46 Countries in Africa have legislation for mandatory fortification of salt

Legislation scope for salt in Rwanda	
Type of Salt that must be fortified	All types (no exceptions)
Origins or destinations of Salt that must be fortified	✔ Domestically produced ✔ Imported ✖ Exports
Intended use of Salt that must be fortified	✔ Household ✔ Processed Food ✔ Animal Feed ✖ Donated food

Source: Rwanda Food and Drugs Authority. Regulations No. CBD/TRG/003 Rev No. 1 Governing Food Fortification in Rwanda. Kigali, Rwanda. 14/January/2020

Nutrient levels and compounds in salt fortification standard in Rwanda		
Iodine	Potassium iodate All compounds are WHO recommended	40 mg/kg 82% of the WHO recommended level

Source: RS EAS 35 - Fortified Salt - Specification

47 Countries in Africa have salt fortification standards

Presence of monitoring protocols for salt fortification in Rwanda	
External monitoring of domestic production	Not applicable
Imported food monitoring	Unknown

Source: Worldatlas. The World's Top Salt Producing Countries. Extracted 22/January/2019. [https://bit.ly/2U9Rgbg], IndexMundi. Salt Production by Country (Thousand metric tons). Extracted 22/January/2019. [https://bit.ly/2MlWKwP], British Geological Survey. Mineral Production 2008 to 2012. Extracted 22/January/2019. [https://bit.ly/2DrALBQ]

Source: IndexMundi. Salt (incl. table salt & denatured salt) & pure sodium chloride, whether or ... Imports by Country in US Dollars. Extracted 20 August 2020. [https://bit.ly/3heCICU], Tridge. Top Importing Countries of Salt. Extracted 20 August 2020. [https://bit.ly/329Zs0D], World's Richest Countries. Salt Importers. Extracted 20 August 2020. [https://bit.ly/3g9M41y]

4 Countries in Africa with mandatory fortification of salt have external monitoring protocols

9 Countries in Africa with mandatory fortification of salt have import monitoring protocols

Wheat flour

Mandatory fortification since 2019

Source: Rwanda Food and Drugs Authority. Regulations No CBD/TRG/003 Rev. No 1 governing Food Fortification in Rwanda, (Rwanda FDA law No. 003/2018 of 09/02/2019, Article 8). Rwanda. 14/January/2020. [http://www.rwandafda.gov.rw/web/fileadmin/RegulationsGoverningFoodFortification.pdf]

31 Countries in Africa have legislation for mandatory fortification of wheat flour

Legislation scope for wheat flour in Rwanda	
Type of Wheat flour that must be fortified	All types (no exceptions)
Origins or destinations of Wheat flour that must be fortified	✔ Domestically produced ✔ Imported ✖ Exports
Intended use of Wheat flour that must be fortified	✔ Household ✔ Processed Food ✔ Animal Feed ✖ Donated food

Source: Rwanda Food and Drugs Authority. Regulations No CBD/TRG/003 Rev. No 1 governing Food Fortification in Rwanda, (Rwanda FDA law No. 003/2018 of 09/02/2019, Article 8). Rwanda. 14/January/2020. [http://www.rwandafda.gov.rw/web/fileadmin/RegulationsGoverningFoodFortification.pdf]

Nutrient levels and compounds in wheat flour fortification standard in Rwanda		
Vitamin A	Retinyl palmitate All compounds are WHO recommended	0.95 mg/kg 16% of the WHO recommended level
Iron	Ferrous fumarate All compounds are WHO recommended	40 mg/kg 67% of the WHO recommended level
Thiamin (B1)	Thiamin mononitrate All compounds are WHO recommended	4.6 mg/kg 153% of the WHO recommended level
Riboflavin (B2)	Riboflavin All compounds are WHO recommended	3.3 mg/kg 165% of the WHO recommended level
Niacin (B3)	Niacinamide All compounds are WHO recommended	30 mg/kg 75% of the WHO recommended level
B6	Pyridoxine hydrochloride All compounds are WHO recommended	3 mg/kg 150% of the WHO recommended level
Folate (B9)	Folic acid All compounds are WHO recommended	2.15 mg/kg 43% of the WHO recommended level
B12	Cyanocobalamin All compounds are WHO recommended	0.01 mg/kg 25% of the WHO recommended level
Zinc	Zinc oxide All compounds are WHO recommended	60 mg/kg 63% of the WHO recommended level
Iron	NaFeEDTA All compounds are WHO recommended	30 mg/kg 75% of the WHO recommended level

Source: Rwanda Standards Board. RS EAS 767 (IDT), Second edition, Adopted by RSB 2019-09-13, Fortified wheat flour-specification, ICS 67.060, Reference number RS EAS 767: 2019. Rwanda. 13/September/2019.

34 Countries in Africa have wheat flour fortification standards

Presence of monitoring protocols for wheat flour fortification in Rwanda	
External monitoring of domestic production	Unknown
Imported food monitoring	Unknown

Source for external monitoring protocols: Not applicable
Source for import monitoring protocols: Not applicable

11 Countries in Africa with mandatory fortification of wheat flour have external monitoring protocols

14 Countries in Africa with mandatory fortification of wheat flour have import monitoring protocols

Foundational document details

For food under mandatory fortification, are these essential details captured in key program documents?

Maize flour | Oil | Rice | Salt | Wheat flour

- Fortification Legislation and Standards
- Labeling
- External Monitoring Protocol(s)
- Import Monitoring Protocol(s)
- Enforcement / Penalties
- Laboratory